

Curriculum vitae



POST APPLIED FOR: Assistant Professor	SUBJECT: Biotechnology
FULL NAME: GYANENDRA SINGH SENGAR	
DATE OF BIRTH: 21-01-1889	CASTE-GEN/OBC/SC/ST: GENERAL
CONTACT NO.: +91-9457266575 ^ +91-7906909482 E- MAIL ID: gsengar71@gmail.com	
PRESENT ADDRESS: Animal Health Section, ICAR-National Research Centre on Pig, Guwahati, Assam	
PERMANENT ADDRESS: H.No.-E24, Airport Enclave, Near B.R. Ambedkar Hawai Strip, Meerut, Uttar Pradesh	
PLEASE TICK (✓) IN THE APPROPRIATE BOX	
☐ MALE ✓	☐ FEMALE
☐ MARRIED ✓	☐ UNMARRIED

ACADEMIC QUALIFICATIONS (STARTING FROM THE HIGHEST DEGREE)

NAME OF THE DEGREE/ EXAM	SUBJECT / SPECIALIZATION	NAME OF SCHOOL/ COLLEGE/UNIVERSITY	BOARD/ UNIV.	YEAR OF PASSING	DIVISION/ GRADE	% OF MARKS/CG PA	REMARKS
Class 10 th	Science	S.V.Mandir, HR.Sec. Sch. Muzaffarnagar	U.P.Board	2004	FIRST	62.33%	-
Class 12 th	Science	Sanatan Dharm Int. Coll. Sadar, Meerut	U.P.Board	2006	FIRST	66.4%	-
B.Sc.	Biotechnology	Bhagwati Coll. of Mang. & Tech, Meerut	Ch. Charan Singh University, Meerut, U.P.	2009	FIRST	70.75%	-
M.Sc.	Biotechnology	Dept. of Biotechnology Ch. Charan Singh University, Campus, Meerut, U.P.	Ch. Charan Singh University, Meerut, U.P.	2011	FIRST	73.28%	-
Ph.D.	Biotechnology (Animal Science)	Sam Higginbottom University of Agriculture, Technology and Sciences, Prayagraj, U.P.	Sam Higginbottom University of Agriculture, Technology and Sciences, Prayagraj, U.P.	2019	FIRST	9.2 (CGPA)	-

ANRF-National Post Doc Fellow (ANRF-NPDF)	Animal Biotechnology	ICAR-National Research Centre on Pig, Guwahati, Assam	Anusandhan National Research Foundation, GOI, New Delhi	2022-2024	-	-	-
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Status of Ph. D. Work

a)	Topic	Elucidation of functional intervention of heat shock protein 70 on expression profile of selected pattern recognition receptors during thermal stress in cattle
b)	University/Institution	Sam Higginbottom University of Agriculture, Technology and Sciences, Prayagraj, U.P.
c)	Name and designation of the Supervisor/Guide	Dr. Rajiv Kant, PhD, Professor
d)	Date of Award of Ph.D.	02-12-2019
e)	Status of Completion, if pursuing	Not Applicable

OTHER QUALIFICATIONS

NET (JRF) ICAR-SRF	YES / NO	If yes, Score & year	2012
SET/SLET/GPAT	YES / NO	If yes, Score & year	168/250, 2012

EXPERIENCE (Starting from the Current/ Last job)

	NAME OF THE ORGANIZATION	DESIGNATION	DURATION (From - To)
Teaching Experience			
	Bhagwati College of Managemnet & Technology, Meerut, U.P.	Lecturer	02-07-2011 to 13 August 2012
Research Experience			
	ICAR-National Research Centre on Pig, Guwahati, Assam	National Post Doctoral Fellow (NPDF)	15-12-2022 to 14-12-2024
	ICAR-National Research Centre on Pig, Guwahati, Assam	Research Associate	17-09-2021 to 14-12-2022
	ICAR-Central Research Centre on Buffaloes, Hisar, Haryana	Research Associate	12-07-2021 to 08-09-2021
	ICAR-Central Institute for Research on Cattle, Meerut, U.P.	Research Associate	09-02-2018 to 31-03-2021

	Project Directorate on Cattle, Meerut, U.P.	Research Associate	20-06-2013 to 31-03-2014
	Project Directorate on Cattle, Meerut, U.P.	Senior Research Fellow	14-08-2012 to 19-06-2013

PUBLICATIONS

1. Number of Publications in Journal: (i) Research Publications (with details):

National: **08**

Sl. No	Author(s)	Title	Name of Journal	Volume/D OI	Page	Year
1.	Niharika, J., Deb, R., Parihar, R., Thakur, P.K., Anjaria, P., Sengar, G.S. , Chaudhary, P., Pegu, S.R., Attupurum, N., Antony, N. and Rajkhowa, S.	Isolation and Characterization of Extended-Spectrum β -Lactamase Producing Escherichia coli from Pig Farms and Slaughterhouse	<i>Indian Journal of Microbiology</i>	https://doi.org/10.1007/s12088-023-01151-z	1-7	2024
2.	Anjaria, P., Deb, R., Sengar, G.S. , Linda, N., Pegu, S.R., Rajkhowa, S., Das, P.J. and Gupta, V.K.,	Development of a Culture-Free Herbal-Aided Method for Rapid Detection of Extended Spectrum β -Lactamase in Piggery Farms and Slaughterhouses	<i>National Academy Science Letters-India</i>		1-4	2024
3.	Sengar, G.S. , Chakravarti, S., Deb, R., Pegu, S.R., Sonowal, J., Rajkhowa, S., Das, P.J. and Gupta, V.K.,	In-house-built isothermal visual assays for rapid detection of African swine fever	<i>Current Science</i>	126(3)	298	2024
4.	Deb, R., Sengar, G.S. , Pegu, S.R., Rajkhowa, S., and Gupta, V.K.	NUCLEOFAST viral DNA isolation kit	<i>Current Science</i>	126(1)	0012	2024
5.	Deb, R., Sengar, G.S. , Chakravarti, S., Linda, N., Pegu, S.R., Rajkhowa, S., Das, P.J. and Gupta, V.K.,	A universal primer set for specific vis-à-vis simultaneous differentiation of Escherichia coli and Klebsiella species	<i>Current Science</i>	125(11)	1157	2023
6.	Singh, R., Deb, R., Singh, U., Raja, T.V., Alex, R., Kumar, S., Chakraborti, S., Alyethodi, R.R., Sharma, S. and Sengar, G.	Heterozygosity at the SNP (rs 136500299) of ITGB6 receptor gene possibly influences the susceptibility among crossbred bull to Foot and Mouth Disease infection	<i>Virus Disease</i>	26 (1-2)	48-54	2015

7.	Deb, R., Singh, U., Kumar, S., Singh, R., Sengar, G. and Sharma, A.	Profiling of bovine breast cancer 1, early onset (BRCA1) gene among Frieswal (HF x Sahiwal) cows and their association with Mastitis.	<i>National Academy Science Letters-India</i>	37(6)	579-583	2014
8	Maity H K, Deb R, Sengar, G S , Pegu S R, Rajkhowa S, Acharya, A P, Das P J, Gupta V K.	Virus-like particle against Indian isolate of porcine circovirus 2d – a viable approach for developing a safe and effective vaccine.	<i>Current Science</i>	127 (7)	787	2024

International: 44

Sl. No	Author(s)	Title	Name of Journal	Volume/D OI	Page	Year
1.	Rajkhowa, S., Sonowal, J., Sengar, G.S. , Pegu, S.R., Deb, R., Das, P.J., Doley, J., Paul, S. and Gupta, V.K.	Assessment of reference genes for qRT-PCR normalization to elucidate host response to African swine fever infection	<i>Brazilian Journal of Microbiology</i>	55	1-10	2024
2.	Sengar, G.S. , Chakravarti, S., Deb, R., Pegu, S.R., Anjaria, P., Sonowal, J., Rajkhowa, S., Das, P.J. and Gupta, V.K.	Comparative assessment of two in-house-built isothermal assays for visual detection of African swine fever virus	<i>Journal of Biosciences</i>	49(3)	1-10	2024
3.	Das, P.J., Sonowal, J., Sengar, G.S. , Pegu, S.R., Deb, R., Kumar, S., Banik, S., Rajkhowa, S. and Gupta, V.K.	Characterization of an African swine fever virus outbreak in India and comparative analysis of immune genes in infected and surviving crossbreed vs. indigenous Doom pigs	<i>Archives of Virology</i>	169(7)	145	2024
4.	Parihar, R., Deb, R., Niharika, J., Thakur, P., Pegu, S.R., Sengar, G.S. , Sonowal, J., Das, P.J., Rajkhowa, S., Raj, A. and Gupta, V.K.,	Development of triplex assay for simultaneous detection of Escherichia coli, methicillin resistant and sensitive Staphylococcus aureus in raw pork samples of retail markets	<i>Journal of Food Science and Technology</i>	169(3)	54	2024
5.	Deb, R., Sengar, G.S. , Sonowal, J., Pegu, S.R., Das, P.J., Singh, I., Chakravarti, S., Selvaradjou, A., Attupurum, N., Rajkhowa, S. and Gupta, V.K.	Transcriptome signatures of host tissue infected with African swine fever virus reveal differential expression of associated oncogenes.	<i>Archives of Virology</i>	193(3)	54	2024
6.	Pegu, S.R., Sonowal, J., Deb, R., Das, P.J., Sengar, G.S. , Rajkhowa, S. and Gupta, V.K.	Clinicopathological and ultrastructural study of African swine fever outbreak in North-East India	<i>Microbial Pathogenesis</i>	185	106452	2023
7.	Rajkhowa, S., Sonowal, J., Pegu, S.R., Sanger,	Natural co-infection of pigs with African swine	<i>Brazilian Journal of Microbiology</i>	10.1007/s42770-023-	1-6	2023

	G.S. , Deb, R., Das, P.J., Doley, J., Paul, S. and Gupta, V.K.,	fever virus and porcine reproductive and respiratory syndrome virus in India		01203-y		
8.	Rajkhowa, S., Sonowal, J., Borthakur, U., Pegu, S.R., Deb, R., Das, P.J., Sengar, G.S. and Gupta, V.K.,	Meta-Analysis of the Prevalence of Porcine Zoonotic Bacterial Pathogens in India: A 13-Year (2010–2023) Study	<i>Pathogens</i>	12(10)	1266	2023
9.	Niharika, J., Thakur, P., *Sengar, G.S. , Deb, R., Parihar, R., Sonowal, J., Chaudhary, P., Pegu, S.R., Das, P.J., Rajkhowa, S. and Gupta, V.K.,	Whole genome sequencing-based cataloguing of antibiotic-resistant genes in piggery waste borne samples	<i>Gene</i>	887	147786	2023
10.	Thakur, P.K., Deb, R., Pegu, S.R., Parihar, R., Niharika, J., Das, P.J., Sengar, G.S. , Sonowal, J., Chaudhary, P., Selvaradjou, A. and Raj, A.,	Characterization of piggery farm waste-borne bacterial transposable elements associated with antimicrobial resistance phenotypes	<i>Comparative Immunology, Microbiology and Infectious Diseases</i>	98(1)	102005	2023
11.	Pegu, S.R., Das, P.J., Sonowal, J., Sengar, G.S. , Deb, R., Yadav, A.K., Rajkhowa, S., Choudhury, M., Gulati, B.R. and Gupta, V.K.,	Japanese Encephalitis Virus Genotype III Strains Detection and Genome Sequencing from Indian Pig and Mosquito Vector	<i>Vaccines</i>	11(1)	150	2023
12.	Deb, R., Yadav, A.K., Sengar, G.S. , Sonowal, J., Lalita, D., Pegu, S.R., Singh, I., Linda, N., Das, P.J., Kumar, S. and Pal, P.,	Development of CD163 receptor-based enzyme-linked immunosorbent assay for diagnosis of porcine reproductive and respiratory syndrome virus	<i>3 Biotech</i>	12(11)	325	2022
13.	Deb, R., Sonowal, J., Sengar, G.S. , Pegu, S.R., Praharaj, M.R., Malla, W.A., Singh, I., Yadav, A.K., Rajkhowa, S., Das, P.J. and Bharati, J	Porcine Circovirus type 2 infected myocardial tissue transcriptome signature	<i>Gene</i>	836	146670	2022
14.	Deb, R., Yadav, A.K., Sengar, G.S. , Pegu, S.R., Paul, S., Rajkhowa, S. and Gupta, V.K.,	Recombinant Antigen-Based Diagnostic Assays of Pig Viral Diseases.	<i>In Protocols for the Diagnosis of Pig Viral Diseases</i>	1	109-126	2022
15.	Pegu, S.R., Deb, R., Das, P.J., Sengar, G.S. , Yadav, A.K., Rajkhowa, S., Paul, S. and Gupta, V.K.	Development of multiplex PCR assay for simultaneous detection of African swine fever, porcine circo and porcine parvo viral infection from clinical samples	<i>Animal Biotechnology</i>	28	1-8	2022
16.	Singh, R., Deb, R., Sengar, G.S. , Raja, T.V., Kumar, S., Singh, U., Das, A.K., Alex, R., Kumar, A., Tyagi, S. and Pal, P.,	Differentially expressed microRNAs in biochemically characterized Frieswal™ crossbred bull semen.	<i>Animal Biotechnology</i>	9	1-14	2021

17.	Deb, R. and Sengar, G. S	Expression pattern of bta-mir-2898 miRNA and their correlation with heat shock proteins during summer heat stress among native vs crossbred cattle	<i>Journal of Thermal Biology</i>	94	https://doi.org/10.1016/j.jtherbio.2020.102771	2020
18.	Deb, R., Sengar, G.S. , Junghare, V., Hazra, S., Singh, U., Alex, R. and Kumar, A.	Characterization of a putative ribosome binding site at the 5' untranslated region of bovine heat shock protein 90	<i>Molecular Biology Reports</i>	47(9)	7061-7071	2020
19.	Singh, U., Alex, R., Kumar, S., Deb, R., Venkatesan Raja, T., Singhal, S., Sengar, G.S. , Singh Rathod, B., Srirama Murthy, K. and Vasant Rao Patil, N.	Association of bovine KISS1 single nucleotide polymorphisms with reproductive traits in Indian Cattle	<i>Reproduction in Domestic Animals</i>	55(8)	922-930	2020
20.	Singh, R., Junghare, V., Hazra, S., Singh, U., Sengar, G.S. , Raja, T.V., Kumar, S., Tyagi, S., Das, A.K., Kumar, A. and Koringa, P.	Database on spermatozoa transcriptogram of categorised Frieswal crossbred (Holstein Friesian X Sahiwal) bulls.	<i>Theriogenology</i>	129	130-145	2019
21.	Singh, I., Deb, R., Kumar, S., Singh, R., Andonissamy, J., Smita, S., Sengar, G.S. , Kumar, R., Ojha, K.K., Sahoo, N.R. and Chandran, R.	Deciphering Foot-and-Mouth Disease (FMD) virus-host Tropism.	<i>Journal of Biomolecular Structure and Dynamics</i>	18	1-13	2019
22.	Sengar, G.S. , Kant, R., Deb, R. and Verma, P.,	Small interfering RNA based knock down of acute heat shock and or GGA inducible bovine heat shock protein 70 may interfere invitro expression pattern of TLR2/4 and NOD1/2.	<i>Journal of Thermal Biology</i>	77	75-85.	2018
23.	Singh, R., Sengar, G.S. , Singh, U., Deb, R., Junghare, V., Hazra, S., Kumar, S., Tyagi, S., Das, A.K., Raja, T.V. and Kumar, A.	Functional proteomic analysis of crossbred (Holstein Friesian X Sahiwal) bull spermatozoa.	<i>Reproduction in Domestic Animals.</i>	53(3)	588-608	2018

24.	Sengar, G.S. , Deb, R., Singh, U., Junghare, V., Hazra, S., Raja, T.V., Alex, R., Kumar, A., Alyethodi, R.R., Kant, R. and Jakshara, S.	Identification of differentially expressed microRNAs in Sahiwal (<i>Bos indicus</i>) breed of cattle during thermal stress.	<i>Cell Stress and Chaperones</i>	23(5)	1019-1032	2018
25.	Sengar, G.S. , Deb, R., Singh, U., Raja, T.V., Kant, R., Sajjanar, B., Alex, R., Alyethodi, R.R., Kumar, A., Kumar, S. and Singh, R.	Differential expression of microRNAs associated with thermal stress in Frieswal (<i>Bos taurus</i> x <i>Bos indicus</i>) crossbred dairy cattle.	<i>Cell Stress and Chaperones</i>	23(1)	155-170	2017
26.	Sengar, G.S. , Deb, R., Raja, T.V., Singh, U., Kant, R., Bhanuprakash, V., Alyethodi, R.R., Kumar, S., Verma, P., Chakraborty, S. and Alex, R.	RT-LAMP assay: an alternative approach for profiling of bovine heat shock protein 70 gene in PBMC cultured model	<i>Molecular Biology Reports.</i>	44(3)	281-287.	2017
27.	Alex, R., Ramesha, K.P., Singh, U., Kumar, S., Alyethodi, R.R., Deb, R., Sharma, S., Sengar, G.S. , Kumar, A. and Prakash, B.	Promoter variants of OAS1 gene are associated with reproductive performance and incidence of normal calving in cattle	<i>Theriogenology</i>	11:108	255-261	2017
28.	Alex, R., Ramesha, K.P., Singh, U., Kumar, S., Alyethodi, R.R., Deb, R., Rai, S., Sharma, S., Sengar, G.S. , Kumar, A. and Prakash, B.	Association analysis of novel polymorphisms in 2',5'-oligoadenylate synthetase gene with reproductive traits in indigenous and crossbred cattle of Indian origin	<i>Reproduction in Domestic Animals</i>	53(2)	442-449	2017
29.	Alyethodi, R.R., Singh, U., Kumar, S., Deb, R., Alex, R., Sharma, S., Sengar, G.S. and Prakash, B.	Development of a fast and economical genotyping protocol for Bovine Leukocyte Adhesion Deficiency (BLAD) in cattle	<i>Springer Plus</i>	5(1)	1442	2016
30.	Deb, R., Sengar, G.S. , Singh, U., Kumar, S., Alyethodi, R.R., Alex, R., Raja, T.V., Das, A.K. and Prakash, B., (#Equally contributed)	Application of a loop-mediated isothermal amplification assay for rapid detection of cow components adulterated in buffalo milk/meat.	<i>Molecular Biotechnology</i>	58	850-860	2016
31.	Deb, R., Sajjanar, B., Singh, U., Kumar, S., Sengar, G.S. , Alex, R., Das, A.K., Tyagi, S., Raja, T.V., Alyethodi, R.R. and Singh, R.,	Cytosine deletion at AP2 box region of <i>HSP70</i> promoter and its influence on semen quality traits in crossbred bulls.	<i>Journal of Genetics</i>	95 (4)	1035-1038	2016

32.	Bhanuprakash, V., Singh, U., Sengar, G.S. , Raja, T.V., Sajjanar, B., Alex, R., Kumar, S., Alyethodi, R.R., Kumar, A., Sharma, A. and Kumar, S.	Comparative expression profile of NOD1/2 and certain acute inflammatory cytokines in thermal stressed cell culture model of native and crossbred cattle.	<i>International Journal of Biometrology</i>	61 (5)	931-941	2016
33.	Alyethodi, R.R., Deb, R., Alex, R., Kumar, S., Singh, U., Tyagi, S., Mandal, D.K., Raja, T.V., Das, A.K., Sharma, S. and Sengar, G.S.	Molecular markers, BM1500 and UMN2008, are associated with post-thaw motility of bull sperm.	<i>Animal Reproduction Science.</i>	174	143-149	2016
34.	Bhanuprakash, V., Singh, U., Sengar, G. , Sajjanar, B., Bhusan, B., Raja, T.V., Alex, R., Kumar, S., Singh, R., Kumar, A. and Alyethodi, R.R.	Differential effect of thermal stress on HSP70 expression, nitric oxide production and cell proliferation among native and crossbred dairy cattle.	<i>Journal of Thermal Biology</i>	59	18-25	2016
35.	Deb, R., Sajjanar, B., Singh, U., Alex, R., Raja, T.V., Alyethodi, R.R., Kumar, S., Sengar, G. , Sharma, S., Singh, R. and Prakash, B.	Understanding the mechanisms of ATPase beta family genes for cellular thermotolerance in crossbred bulls,	<i>International Journal of Biometrology</i>	59	1783-1789	2015
36.	Deb, R., Singh, U., Raja, T.V., Kumar, S., Tyagi, S., Alyethodi, R.R., Alex, R., Sengar, G. and Sharma, S.	Designing of an Artificial Neural Network model to evaluate the association of three combined Y specific microsatellite loci on the actual and predicted post thaw motility in crossbred bull semen	<i>Theriogenology</i>	83 (9)	1445–1450	2015
37.	Kumar, S., Deb, R., Singh, U., Ganguly, I., Mandal, D.K., Tyagi, S., Kumar, M., Sengar, G. , Sharma, S., Singh, R. and Singh, R	Bovine Circadian Locomotor Output Cyclic Kaput (CLOCK) and Clusterin (CLU) mRNA quantitation in ejaculated crossbred bull spermatozoa	<i>Reproduction in Domestic Animals</i>	50:(3)	505-509	2015
38.	Singh, R., Deb, R., Singh, U., Raja, T.V., Alex, R., Kumar, S., Chakraborti, S., Alyethodi, R.R., Sharma, S. and Sengar, G.	Heterozygosity at the SNP (rs 136500299) of ITGB6 receptor gene possibly influences the susceptibility among crossbred bull to Foot and Mouth Disease infection	<i>Virus Disease</i>	26 (1-2)	48-54	2015

39.	Singh, R., Deb, R., Singh, U., Alex, R., Kumar, S., Chakraborti, S., Sharma, S., Sengar, G. and Singh, R.,	Development of a tetra-primer ARMS PCR-based assay for detection of a novel single-nucleotide polymorphism in the 5'untranslated region of the bovine ITGB6 receptor gene associated with foot-and-mouth disease susceptibility in cattle	<i>Archives of Virology</i>	159 (12)	3385-3389	2014
40.	Kumar, S., Deb, R., Singh, U., Ganguly, I., Mandal, D.K., Singh, R., Sharma, S., Sengar, G. , Singh, R., Kumar, M. and Sharma, A.	SNPs at exonic region of Aquaporin-7 (AQP-7) gene may affect semen quality parameters among crossbred bulls.	<i>Journal of Genetics</i>	94 (1)	e108 - e112	2014
41.	Deb, R., Sajjanar, B., Singh, U., Kumar, S., Singh, R., Sengar, G. and Sharma, A.,	Effect of heat stress on the expression profile of Hsp90 among Sahiwal (<i>Bos indicus</i>) and Frieswal (<i>Bos indicus</i> X <i>Bos taurus</i>) breed of cattle: A comparative study.	<i>Gene</i>	536 (2)	435-40	2014
42.	Deb, R., Singh, U., Kumar, S., Kumar, A., Singh, R., Sengar, G. , Mann, S. and Sharma, A.	Genotypic to Expression Profiling of Bovine Calcium Channel, Voltage-Dependent, Alpha-2/Delta Subunit 1 Gene, and Their Association with Bovine Mastitis Among Frieswal (HF X Sahiwal) Crossbred Cattle of Indian Origin	<i>Animal Biotechnology</i>	25 (2)	128-138	2014
43.	Deb, R., Sajjanar, B., Singh, U., Kumar, S., Brahmane, M.P., Singh, R., Sengar, G. and Sharma, A.,	Promoter variants at AP2 box region of Hsp70.1 affect thermal stress response and milk production traits in Frieswal cross bred cattle.	<i>Gene</i>	15: 532(2)	230-5	2013
44.	Deb, R., Kumar, S., Singh, U., Tyagi, S., Mandal, D.K., Sengar, G. , Singh, R., Kumar, M. and Sharma, A.,	Evaluation of three bovine Y specific microsatellite loci as an alternative biomarker for semen quality traits in crossbred bull.	<i>Animal Reproduction Science</i>	1.605	30; 142 (3-4):121 - 5.	2013

(ii) Articles/other Publications:

Technical articles

- Asish, **Sengar G**, Singh R, Singh U, Deb R (2015) Common pathogens involved for causing bovine mastitis. *Indian Farming* (ISSN: 0019-4786), 65(01).
- Deb R, **Sengar G S**, Pegu S R, Rajkhowa S, Gupta V K (2024) Beneath the Surface: Pig as unseen carriers of Rat Hepatitis Virus and their impact on human health. *Bione*. Published By Babrone Team, College of Veterinary Science, Khanapara, Guwahati-22, 781022. <http://babrone.edu.in/blog/?p=5677>.
- Deb R, Deb M, **Sengar G S**. (2024) Navigating Metrics: Maximizing Research Impact in Academic Publishing. *Trends Scholarly Publishing.*, 3 (1): 10-13, 2024. <https://doi.org/10.17311/tsp.2024.10.13>.
- Kar, P. Anjaria, P., **Sengar, G.S.**, Pegu, S.R., Deb, R. and Gupta, V.K. (2023) Advancing Swine Health: Field-Based Perspectives on Pen-Side Diagnostic Tools. *Food and Scientific Reports*, 4(12): 39-43.
- देब. राजिब., अंजारिया. प्रनव., पेगु. सीमा. रानी., सेंगर. ज्ञानेन्द्र सिंह, गुप्ता. विवेक., (२०२४) अफ्रीकन स्वाइन फीवर. विज्ञान प्रगति. अगस्त २०२४. पृष्ठ संख्या-३३-३४

2. Number of Books written and published: 05

1. **Sengar G**, Deb R, Singh U (2016). Polymerase Chain Reaction (PCR): Principle to practices. Published by Today and Tomorrow's Printers and Publishers, New Delhi, India **ISBN: 81-7019-544-2**, 9788170195443, **Pages 218 nos.**
2. Deb R, Patil NV, **Sengar GS** (2020) HSP70s Discovery, structure and functions. Published by Nova Science Publishers, Inc., New York. **ISBN:978-1-53618-179-1. Total Page nos.175.**
3. Singh R, **Sengar G**, Deb R (2016) Monograph on Genetic Basis of Host Receptor on Foot and Mouth Disease: FMDV host receptor. **ISBN 9783659922794**. LAP Lambert Academic Publishing, OmniScriptum GmbH &Co. KG, Bahnhofstraße, 28, D-66111, Saarbrücken, Germany. 80 Pages.
4. Bhanuprakash V, **Sengar G**, DebR (2016) Immune-Thermal Crosstalk in Cattle. **ISBN 659923567**. LAP Lambert Academic Publishing, OmniScriptum GmbH &Co. KG, Bahnhofstraße, 28, D-66111, Saarbrücken, Germany. 80 Pages.
5. Deb R, **Sengar G S**, Gupta V K (2025) Emerging Zoonotic Threats from Swine. Springer Nature Publisher (Accepted)

3. Number of Book chapters contributed: 15

Deb R, Mukhopadhyay C S, **Sengar GS**, Cruz A S da, Silva D C, Pinto I P, Minasi L, Aleves Costa E O, Cruz A D. da (2019) Genetic markers for improving farm animals . In *Genomics and Biotechnological advances in veterinary, poultry and Fisheries* edited by Malik YS, Barh D, Azevedo V and Khurana S M Paul. Academic Press and Elsevier. ISBN 978-0-12-816352-8. Page no. 107-129.

Sengar G S, Deb R, Chakraborty S, Mandal K, Bhanuprakash V, Singh, U. 2016. Overview of Pain in livestock: Mechanism to nutritional control 2016 in *Nutritional Modulators of Pain in the Aging Population* Edited by Ronald Ross Watson and Sherma Zibadi Elsevier and Academy Press ISBN: 9780128051863.

Deb R, Yadav A K, **Sengar G S**, Pegu S R, Paul S, Rajkhowa S and Gupta V K (2022) Recombinant antigen-based diagnostic assays of pig viral diseases. In Rajib Deb et al. (eds.), *Protocols for the Diagnosis of Pig Viral Diseases*, Springer Protocols Handbooks, https://doi.org/10.1007/978-1-0716-2043-4_1

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Deb R, **Sengar G S**. 2018 Cloning of DNA fragment in E coli. In Genetic Engineering and Cytogenetics tools: hand guided student practical manual on basic genetic engineering, molecular and cytogenetics tools. Goiania:/Editora Academico, Brazil. 2018. ISBN:978-85-5440-132-0.

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4. Any other information:

Intellectual Property Rights

Patent (Granted) :02

1. Deb, R., Alex, R., Kumar, S., Singh, U., Raja, T.V., Alyethodi, R.R., **Sengar, G.**, Prakash, B. “A color-based

assay for differentiating Cow vs Goat milk/meat". Patent granted on 24.01.2024 by The Patent Office, Government of India. (**Patent Number # 503089**).

2. Deb, R., Singh, U., Kumar, S., Das, A.K., Raja, T.V., Alyethodi, R.R., Alex, R., **Sengar, G.**, Prakash, B. "Development of an in-house built lamp assay for rapid detection of Cow components adulterated in Buffalo milk/meat". Patent granted on 29.11.2023 by The Patent Office, Government of India. (**Patent Number # 474562**).

Patent (Filed): 07

1. **Sengar, G.S.**, Deb, R., Chakarvarti, S., Pegu, S.R., Rajkhowa, S., Das, P.J., Gupta, V.K. "A provisional Indian patent application entitled "Novel Primers for isothermal amplification vis a vis visual rapid detection and quantitative detection of African swine fever viral DNA " Indian Patent filled with application number **202311029459** on 24-04-2023 by The Patent Office, Government of India.
2. **Sengar, G.S.**, Deb, R., Pegu, S.R., Das, P.J., Gupta, V. K. (2024) Primers, peptides and antibody for development of mango leaves derived nanoparticle based lateral flow assay for diagnosis of African swine fever virus"- Filed for Indian Patent application (complete specification), dated 5th Dec, 2024
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4. Deb, R., **Sengar, G.S.**, Chakarvarti, S., Ningthoukhongjam, L., Pegu, S.R., Rajkhowa, S., Das, P.J., Gupta, V.K." Novel universal single set of primers for specific vis-à-vis simultaneous differentiation of Escherichia coli and Klebsiella species" Indian Patent filled with application number **202311041898** on 23-06-2023 by The Patent Office, Government of India.
5. Deb, R., Kumar, A., **Sengar, G.S.**, Kumar, S., Raja, T.V., Singh, U., Kumar, S., Patil, N.V." Kit for differentiating cow and camel milk." Indian Patent filled with application number **201911031389** on 27-08-2021 by The Patent Office, Government of India.
6. Deb, R., Singh, U., **Sengar, G.S.**, Sajjanar, B., Kumar, S., Hazara, S., Alex, R., Raja, T.V., Alyethodi, R.R., Kumar, A., Prakash, B." Identification of functional Internal Ribosomal Entry Sites (IRES) at bovine Heat Shock Protein 90." Indian Patent filled with application number **201711042562** on 11-01-2019 by The Patent Office, Government of India.
7. Alyethodi, R.R., Singh, U., Kumar, S., Alex, R., Deb, R., Raja, T.V., **Sengar, G.S.**, Prakash, B. "Development of tetra primer displacement PCR assay for detection of complex vertebral malformation in cattle." Indian Patent filled with application number **201611034916** on 13-04-2018 by The Patent Office, Government of India.

Copyrights (Granted): 03

1. Pegu S R, Deb R, Das P J, **Sengar G**, Rajkhowa S, Gupta V K (2023) Piggyplex (D) ASFV, PCV &PPV assay kit -copyright registered with certificate no. L135409/2023, dated 03.11.2023
2. Pegu S R, Deb R, Das P J, **Sengar G**, Rajkhowa S, Gupta V K (2023) Piggyplex ® CSF, JE & PRRS assay kit Copyright registered with certificate no. L142041/2024, dated 25.01.2024
3. Deb R, **Sengar G**, Pegu S R, Rajkhowa S, Gupta V K (2023) Nucleofast Viral DNA isolation kit Copyright registered with certificate no. L135405/2023, dated 03.11.2023

Trademarks (Granted): 03

1. Pegu S R, Deb R, Das P J, **Sengar G**, Rajkhowa S, Gupta V K (2023) Piggyplex D. Trademark registered with certificate no. 3302025, dated 10.11.2023
2. Pegu S R, Deb R, Das P J, **Sengar G**, Rajkhowa S, Gupta V K (2023) Piggyplex R. Trademark registered with certificate no. 3305191, dated 12.11.2023
3. Deb R, **Sengar G**, Pegu S R, Rajkhowa S, Gupta V K (2023) Nucleofast. Trademark registered with certificate no. 3302090, dated 10.11.2023

Technologies developed

Sl No	Name of the technology/product s/new initiatives	Brief description of the technology	Developers	Clickable URL/links /documentary proofs
1	CIRC-COWCAM assay Kit/ Detection of adulteration of camel milk with cow milk or vice-versa	The CIRC-COWCAM assay kit is useful for detecting cow milk adulteration in camel milk as well and vice versa at a minimum level of 1%. The test can be completed less than 4 hours. The kit is less expensive, user friendly, rapid and specific in detection of adulteration of camel milk with cow milk and vice-versa. It will be useful for the dairy industry, NGOs and other stakeholders dealing with the sale of camel / cow milk. Technology Details: The CIRC-COWCAM Assay kit is useful for detecting cow milk adulteration in camel milk as well as vice versa at a minimum level of 1%. The test can be completed less than 4 hours. The kit is less expensive, user friendly, rapid and specific in detection of adulteration of camel milk with cow milk or vice-versa. It will be useful for the dairy industry, NGOs and other stakeholders dealing with the sale of camel / cow milk.	Rajib Deb, Amod Kumar, T V Raja, Umesh Singh, Sushil Kumar, N V Patil, Gyanendra S Sengar	https://circ.icar.gov.in/patenttechnologies.php https://biospub.com/index.php/biorestoday/article/download/41/33 https://www.istem.gov.in/digitalcatalogue/showinfo/TP19769653432 http://agrinnovateindia.co.in/subTechnologyDescription?id=306&Subtitle=CIRC-COWCAM https://icar.org.in/node/12312 (Release of the Kit from ICAR) https://icar.org.in/sites/default/files/2024-03/ICAR-AR-2020.pdf (Page no.102) https://sansad.in/getFile/annex/259/AU2721.pdf?source=pqars (Page no 2- 4C-Vii)

2	Development of an in-house built LAMP assay for rapid detection of cow components adulterated in buffalo milk/meat	An in-house built rapid, economical and user-friendly LAMP assay protocol to identify the buffalo milk/meat samples adulterated with cow milk/meat was developed. This test can be completed within 1 hour 45 minutes and can detect upto 5% adulterated cow DNA in buffalo counter parts.	Rajib Deb, Umesh Singh, Sushil Kumar, T V Raja, Rafeeqe R. Alyethodi, Rani Alex, Gyanendra Singh and B Prakash	https://krishi.icar.gov.in/Technology/technologyShowCaseReport.jsp?id=201521198075258 https://icar.org.in/patent-granted-icar-circ-meerut-detection-adulteration-buffalo-milk-or-meat-cow-milk-or-meat https://circ.icar.gov.in/patenttechnologies.php https://link.springer.com/article/10.1007/s12033-016-9984-4 (Research Paper Published)
3	A visible test for differentiating cow and goat milk/meat	A post PCR color based differentiation of cow vs goat milk and meat samples has been developed. The test does not require downstream PCR processing and handling of carcinogenic dyes. Moreover, it is rapid and can be completed within 2-3 hour including genomic extraction from milk/meat samples. The test can detect 2% adulterated milk or 1 mg meat from either cow or goat sources.	Rajib Deb, Rani Alex, Sushil Kumar, Umesh Singh, T V Raja, Rafeeqe R. Alyethodi, Gyanendra Singh and B Prakash	https://krishi.icar.gov.in/Technology/DetailReport.jsp?id=201521151035364 https://circ.icar.gov.in/patenttechnologies.php https://pubmed.ncbi.nlm.nih.gov/28775755/ (Research paper published)
4	CIRC-Cattle CVM assay/Development of tetra primer displacement PCR assay for detection of complex vertebral malformation (CVM) carriers in cattle	CIRC-Cattle CVM assay is used for early detection of CVM carrier animals. The assay may prove to be an effective tool for selecting CVM free animals especially young bulls at their early stage of life. This assay will be useful for the semen collection/germplasm centers, research institutions, State Animal Husbandry Departments and NGOs involved in semen collection and	R.R. Alyethodi, Umesh Singh, Sushil Kumar, Rani Alex, Rajib Deb, Raja T V, Gyanendra S Sengar , B. Prakash	https://krishi.icar.gov.in/Technology/technologyShowCaseReport.jsp?id=201587623859145 https://circ.icar.gov.in/patenttechnologies.php http://agrinovateindia.co.in/subTechnologyDescription?id=317&SubTitle=CIRC-Cattle%20CVM%20assay

		processing of cattle and buffaloes.		
5	Identification of functional internal ribosomal entry sites (IRES) at bovine heat shock protein 90/Bovine heat shock protein 90 internal ribosomal entry sites (IRES) based expression vector	A putative IRES was identified in bovine heat shock protein gene for the first time, which was showed to be functional. The identified bovine heat shock protein IRES was used to develop an artificial expression-cassettes for simultaneous translation of two gene from a same reading frame. An Indian patent has been filed with application number 201711042562 dated 28/11/2017. Patent has been filed and published (The Patent Office Journal no. 02/2019, dated 11/01/2019, Page No: 1129).	Rajib Deb, Umesh Singh, Gyanendra Singh Sengar , Basavraj Sajjanar, Sushil Kumar, Saugata Hazra, Rani Alex, T V Raja, Rafeeqe Rahman Alyethodi, Ashish Kumar, A K Das, B Prakash.	https://circ.icar.gov.in/patenttechnologies.php https://pubmed.ncbi.nlm.nih.gov/32888122/ (Research paper published)
6	Nucleo ^{Fast} Viral DNA isolation kit	An indigenous viral DNA isolation kit, viz. NUCLEO ^{FAST} viral DNA isolation kit has been developed which can isolate ready-to-use template viral DNA for routine PCR-based diagnosis of animal viral diseases without any PCR inhibitory effects. An Indian IP has been filed with application no. NUCLEOFAST105 7478477/207/ASAT MA-1276/ICAR-P.	Rajib Deb, Gyanendra Singh Sengar , Seema Rani Pegu, Swaraj Rajkhowa and Vivek Kumar Gupta	https://krishi.icar.gov.in/Technology/technologyShowCaseReport.jsp?id=201642079957628 https://www.currentscience.ac.in/Volumes/126/01/0012.pdf Current Science (00113891), 2024, Vol 126, Issue 1, p12 (Publication)
7	Single tube tetra ARMS PCR-based assay for genotyping of FMD virus receptor in cattle	A single tube tetra ARMS PCR-based assay for rapid genotyping of the 5' UTR region of bovine Integrin	Rajib Deb, Rani Singh, Umesh Singh, Rani Alex, Sushil Kumar, Soumendu Chakraborti, Sheetal Sharma, Gyanendra Sengar ,	https://krishi.icar.gov.in/Technology/DetailReport.jsp?id=201600080837673 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4310182/

		beta6 receptor gene, linked with Foot and Mouth diseases virus (FMDV) host tropism, has been developed. Four sets of primers were employed: Outer Forward, Outer Reverse, Inner Forward and Inner Reverse. They were designed to amplify fragments of Single tube tetra ARMS PCR-based assay for genotyping FMD virus receptor in cattle different sizes for each allele band; in the order to be easily resolved using agarose gel electrophoresis. To increase specificity of the reaction, a mismatch was introduced at the 3' end of each of the two allele-specific primers.	Rupali Singh and Birham Prakash	https://pubmed.ncbi.nlm.nih.gov/25078391/ (Research Publications)
8	Piggyplex (D) ASF, PCV2 & PPV assay kit	Simultaneous diagnosis of three DNA viruses viz. African Swine Fever Virus (ASFV), Porcine Circovirus (PCV) & Porcine Parvo virus (PPV) in swine blood and tissue samples	Seema Rani Pegu, Rajib Deb, Pranab Jyoti Das, Gyanendra Singh Sengar, S Rajkhowa, V K Gupta	https://krishi.icar.gov.in/Technology/technologyShowCaseReport.jsp?id=201663933691303 http://agrinnovateindia.co.in/subTechnologyDescription?id=890&Subtitle=Piggyplex(D)%20ASFV,%20PCV%20and%20PPV%20assay%20kit
9	Piggyplex ® CSF, JE & PRRS assay kit/ Multiplex CSF, JE & PRRS assay Kit	Classical Swine Fever (CSF), Japanese Encephalitis (JE), and Porcine Respiratory and Reproductive Syndrome (PRRS) are three major transboundary pig diseases that cause significant economic losses for pig farmers. To detect CSF, JE, and PRRS in porcine samples, a	Seema Rani Pegu, Rajib Deb, Pranab Jyoti Das, Gyanendra Singh Sengar, S Rajkhowa, V K Gupta	https://krishi.icar.gov.in/Technology/DetailReport.jsp?id=201646848725695 http://agrinnovateindia.co.in/subTechnologyDescription?id=891&Subtitle=Piggyplex%C2%AE%20CSF,%20JE%20and%20PRRS%20assay%20kit

		specific set of oligonucleotide primers was designed and combined in a multiplex arrangement. To demonstrate the accuracy of the developed multiplex PCR assay, 413 porcine post mortem tissue samples were tested. Each of the CSFV, JEV, and PRRSV target genes was amplified using the designed primers, but no other porcine virus genes were detected. The detection limit of the assay was 10^2 - 10^3 copies/ μ l of viral nucleic acid.		
10	qPCR ^{TP} assay for detection of African Swine Fever Virus	Hydrolysis probe based quantitative assay developed for detection of African swine Fever viral infection in pigs. The developed assay can be able to detect the ASF viral load in clinical samples within 30 minutes (post nucleic acid extraction). The assay is specific for ASFV without any cross-reactivity with other swine viral pathogens. The assay was found to detect as low as 5 femtograms of ASF viral DNA.	Seema Rani Pegu, Gyanendra S. Sengar , Rajib Deb, Pranab Jyoti Das, Swaraj Rajkhowa, Vivek Kumar Gupta	http://agrinnovateindia.co.in/subTechnologyDescription?id=888&Subtitle=qPCR%20TP%20assay%20for%20detection%20of%20African%20Swine%20Fever%20Virus
11	qPCR ^{TP} assay for detection of Porcine Reproductive and Respiratory Syndrome Virus	A quantitative assay developed for detecting Porcine Reproductive and Respiratory Syndrome Virus	Seema Rani Pegu, Gyanendra S. Sengar , Rajib Deb, Pranab Jyoti Das, Swaraj Rajkhowa, Vivek Kumar Gupta	Supportive document Institutional technology code: 300/ITMU/NRCP/2022-23/Technology-33 , dated 13.10.2022 (Attached)
12	Triplex assay for simultaneous detection of <i>Escherichia coli</i> , Methicillin Resistant and Sensitive	Simultaneous detection of three important pork meat borne bacterial pathogens namely <i>Escherichia coli</i> ,	Rajib Deb, Ranjeet Parihar, Seema Rani Pegu, Gyanendra S. Sengar , Swaraj Rajkhowa, Pranab Jyoti	https://link.springer.com/article/10.1007/s13197-023-05917-7 (Research Publication)

	<i>Staphylococcus aureus</i> in raw pork samples	Methicillin Resistant and Sensitive <i>Staphylococcus aureus</i>	Das, Vivek Kumar Gupta	
13	Loop-Mediated Isothermal Amplification (LAMP) and Polymerase Spiral Reaction (PSR) based rapid visual assays for rapid detection of African Swine Fever virus/ In-house-built isothermal visual assays for rapid detection of African swine fever	Two in-house-designed iso-thermal amplification-based visual tests, LAMP (loop mediated isothermal amplification) and PSR (polymerase spiral reaction), was developed for quick and sensitive detection of ASFV viral DNA in porcine clinical samples. The analytical sensitivity for LAMP and PSR was 50 pg and 50 fg, respectively. A provisional Indian patent application was filed on 24 April 2023, with application number 202311029459.	Rajib Deb, Gyanendra S. Sengar , Seema Rani Pegu, Soumendu Chakravarti, Swaraj Rajkhowa, Pranab Jyoti Das, Vivek Kumar Gupta	https://www.currentscience.ac.in/Volumes/126/03/0298.pdf (Research Publication)
14	qPCR assay for detection of Classical Swine Fever Virus	Hydrolysis probe based quantitative assay for detection of Classical Swine Fever virus copy numbers	Seema Rani Pegu, Gyanendra S. Sengar , Rajib Deb, Pranab Jyoti Das, Swaraj Rajkhowa, Vivek Kumar Gupta	http://www.agrinovateindia.co.in/subTechnologyDescription?id=889&Subtitle=qPCR%20assay%20for%20detection%20of%20Classical%20Swine%20Fever%20Virus&Techtitle=ICAR-National%20Research%20Centre%20on%20Pig%20(NRCP)
15	Virus like particle based vaccine candidate for Porcine circo virus 2d	Developed virus like particle-based subunit vaccine candidate against Indian isolate of Porcine circo virus 2d	Hemanta Kumar Maity, Rajib Deb, Gyanendra Singh Sengar , Seema Rani Pegu, Swaraj Rajkhowa, Aditya Pratap Acharya, Pranab Jyoti Das and Vivek Kumar Gupta	Research News (Current science): https://www.currentscience.ac.in/Volumes/127/07/0787.pdf
16	AMR ^{Tester} : Culture free herbal aided antimicrobial resistant tester in livestock products, biological samples and farm waste	Herbal aided diagnostic for rapid detection of antimicrobial resistant bacteria	Rajib Deb, Seema Rani Pegu, Gyanendra Singh Sengar , Nitin M Attupuram, N Linda, S Rajkhowa, P J Das, V K Gupta	Technology Inventory, ICAR-NRCP- Technology code39

International Conference presentations

1. Oral presentation on “Development and comparison of rapid visual detection assays for African swine fever virus (ASFV) based on loop-mediated isothermal amplification (LAMP) and polymerase spiral reaction

(PSR)''' at the 2nd International Workshop on Molecular Diagnostics in Microbiology and Disease (MDMD2023), held in **Quy Nhon, Vietnam, from 11-14th December 2023**

2. Poster presentation on " A duplex TaqMan based quantitative assay for simultaneous detection of African Swine Fever Virus and Porcine Circovirus 2" at the 16th International Conference on Molecular Epidemiology and Evolutionary Genetics of Infectious Diseases (MEEGID XVI) held in **Dresden, Germany, from 14-17th November 2023**
3. Oral presentation on "Cataloguing of novel miRNAs during thermal stress in cattle" at the 3rd International Conference on Transcriptomics in **Bangkok, Thailand, from 30-31st October 2017**

Acted as Reviewer:

PLOS One, Heliyon, International Journal of Biological Macromolecules, BMC Genomics, Transboundary and Emerging Diseases, Journal of Thermal Biology, Journal of Biomolecular Structure and Dynamics, Reproduction, Fertility and Development, 3 Biotech, Reproduction in Domestic Animals, International Journal of Advances in Biology, Molecular Reproduction and Development, Journal of Veterinary Science & Technology, Reproduction, Fertility and Development, International Journal of Advances in Biology (IJAB), Gene Reports

Professional Recognition/ Award/ Prize/ Certificate, Fellowship received

S.No	Name of Award	Awarding Agency	Year
1.	ICAR Senior Research Fellowship Award for PhD	Indian Council of Agricultural Research, New Delhi, India	2013
2.	Young Scientist Award	Indian association of advancement in veterinary Research, Hebbal, Bangalore	2013
3.	Late Dr. M. N. Kulkarni Memorial Award	Indian Association of Veterinary Immunology and Microbiology, Department of Animal Husbandary, Lucknow, UP.	2013
4.	First prize in poster presentation	National seminar on "Integrated development of vast biodiversity of indigenous livestock for long time rural livelihood security" (Feb 7-8, 2013), G.B.Pant univ. of Agri and tech, Pantnagar	2013
5.	First prize in poster presentation	National Seminar on "Technological and Policy Interventions for Sustainable Cattle Breeding in India" organized by Project Directorate on Cattle (ICAR), Meerut in collaboration with the Society for Conservation of Domestic Animal Biodiversity, Karnal (14 th March, 2013)	2013
6.	Second prize in poster presentation	National Symposium on Policy Planning for Livestock security through domestic animal biodiversity and XII annual convention of Society for conservation of Domestic animals' biodiversity, Feb 11-12, 2016 at Sher-e-Kashmir university of Agricultural Sciences and technology of Jammu, Jammu (J&K)-181 102, India	2016
7.	Second prize in oral presentation	International Conference on Advancing Frontiers in Biotechnology for sustainable Agriculture and Health, at Sam Higginbottom Institute of agriculture, technology & Sciences, Allahabad, India 25-26th Feb, 2016.	2016
8.	First prize in poster presentation	International conference on Global initiatives in applied sciences and greentechnologies from 09-11 Sep, 2016 at SRM University, Delhi-NCR campus, Modinagar, Ghaziabad (U.P.), India.	2016

9.	International Travel Grant Award (For Thailand)	Science and Engineering Research Board, DST, Govt of India	2017
10.	Certificate of Appreciation	Received for delivering lectures during ITEC training Program at The Energy and Resources Institute (TERI), India Habitat Centre, Lodhi Road, New Delhi 110003 on 8th Dec, 2016.	2016
11.	Best Poster Award	World Congress on Biotechnology and Biological studies, New Delhi, India	2017
12.	Certificate of Appreciation	Received for delivering lectures during ITEC training Program at The Energy and Resources Institute (TERI), India Habitat Centre, Lodhi Road, New Delhi 110003 on 1st Dec, 2017.	2017
13.	Certificate of Reviewing (Gene Reports)	Elsevier	2017
14.	Certificate of Reviewing (Journal of Thermal Biology)	Elsevier	2018
15.	Certificate of Reviewing (Animal Reproduction Science)	Elsevier	2018
16.	International Travel Grant Award (For Dubai)	Asian Council of Science Editor (ACSE)	2018
17.	Best Doctoral Research Award	Society for Biotic & Environmental Research, Tripura	2020
18.	Best Paper presentation Award	IAAVR	2023
19.	International Travel Grant Award for (For Germany)	Science and Engineering Research Board, DST, Govt of India	2023
20.	ICGEB Grant (For Vietnam)	International Centre for Genetic engineering & Biotechnology, Trieste (Italy),	2023
21.	India Bioscience Award For attending Young Investigator Meeting 2024	India Bioscience, Bengaluru	2024

Total Citation & H-index: 960 & 16 (<https://scholar.google.com/citations?user=eDTEooIAAAAJ&hl=en>)



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